



United International University

School of Science and Engineering

Department of Computer Science and Engineering

Final Exam Trimester:- Spring 2026

Course Title: Fundamental Calculus

Course Code: MATH-1151 Marks:- 40 Time: 2 hours

[Note that the marks are given in brackets at the end of each question or part question. You have to answer all the questions in order.]

1. (a) During a FIFA World Cup match, the distance covered by a football after being kicked is modeled by, $s(t) = 4t^3 - 18t^2 + 24t$. Where, s is the distance in meters and t is the time in seconds.
 - i) Find the velocity of the ball at $t = 2$. [2]
 - ii) Find the acceleration of the ball at $t = 1$. [2]
- (b) Find the coordinates of the turning point of the function,
 $y = \frac{2}{3}x^3 - x^2 - 4x + 9$. Also state their nature. [4]
2. (a) Find the gradient of $y = e^x + 3x - 2$ at $x = 0$.
Hence, Find the equation of the tangent line and normal line at $x = 0$. [5]
- (b) Draw the derivative curve function of the following function: [3]

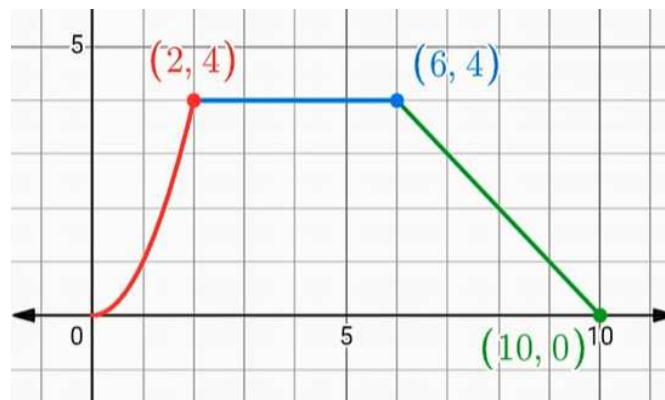


Figure 1:

P.T.O.

3. (a) Given, $\frac{dy}{dt} = \frac{9}{\sqrt{9t-5}}$ and $y = 2$ when $t = 1$.
Find y in terms of t . [4]

- (b) Differentiate the following with respect to x (Any two): [4]

i) $y = \ln \sqrt{\frac{x^2}{1+x^2}}$

ii) $y = xe^{-x} - 9x^2$

iii) $y = \frac{\sin x}{1+\cos x}$

4. (a) Integrate the following (Any two): [4]

i) $\int \cos^3 x \sin x dx$

ii) $\int_1^3 \frac{t^4+t^2+3}{t^2} dt$

iii) $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$

- (b) Evaluate $\int_0^1 t^2 e^t dt$. [4]

5. (a) The following figure shows the enclosed area between two curves $y = \sin x$ and $y = \cos x$.

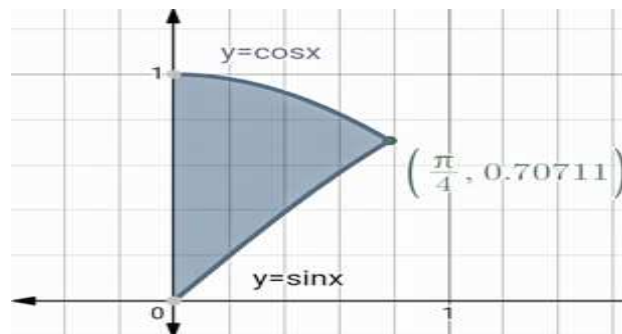


Figure 2:

- Find the area of the shaded region. [3]

- (b) The region under the curve $y = x^3$ over the interval $[0, 1]$ is shown as follows:

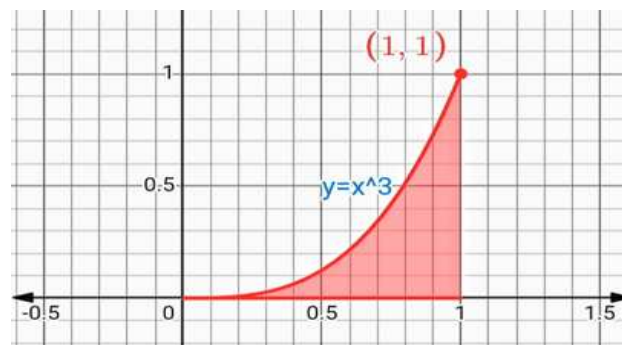


Figure 3:

- i) Find the area of the shaded region. [2]

- ii) If the region is rotated about the x axis by 360° , then find the volume of the resulting solid. [3]